

Diabetic Ketoacidosis



Prepared by

Assist professor Dr \Afaf Mohamed Mansour

Outline

- Definition of DKA
- Comparison of Diabetic Ketoacidosis and Hyperglycemic Hyperosmolar Syndrome
- Etiology of DKA
- Pathophysiology of DKA
- Clinical Manifestation of DKA
- Laboratory Studies
- Management of DKA

Definition of DKA

Diabetic Ketoacidosis is a life-threatening hyperglycemic crisis associated with diabetes mellitus result when glucose cannot transport to the cells because **a lack of insulin.**

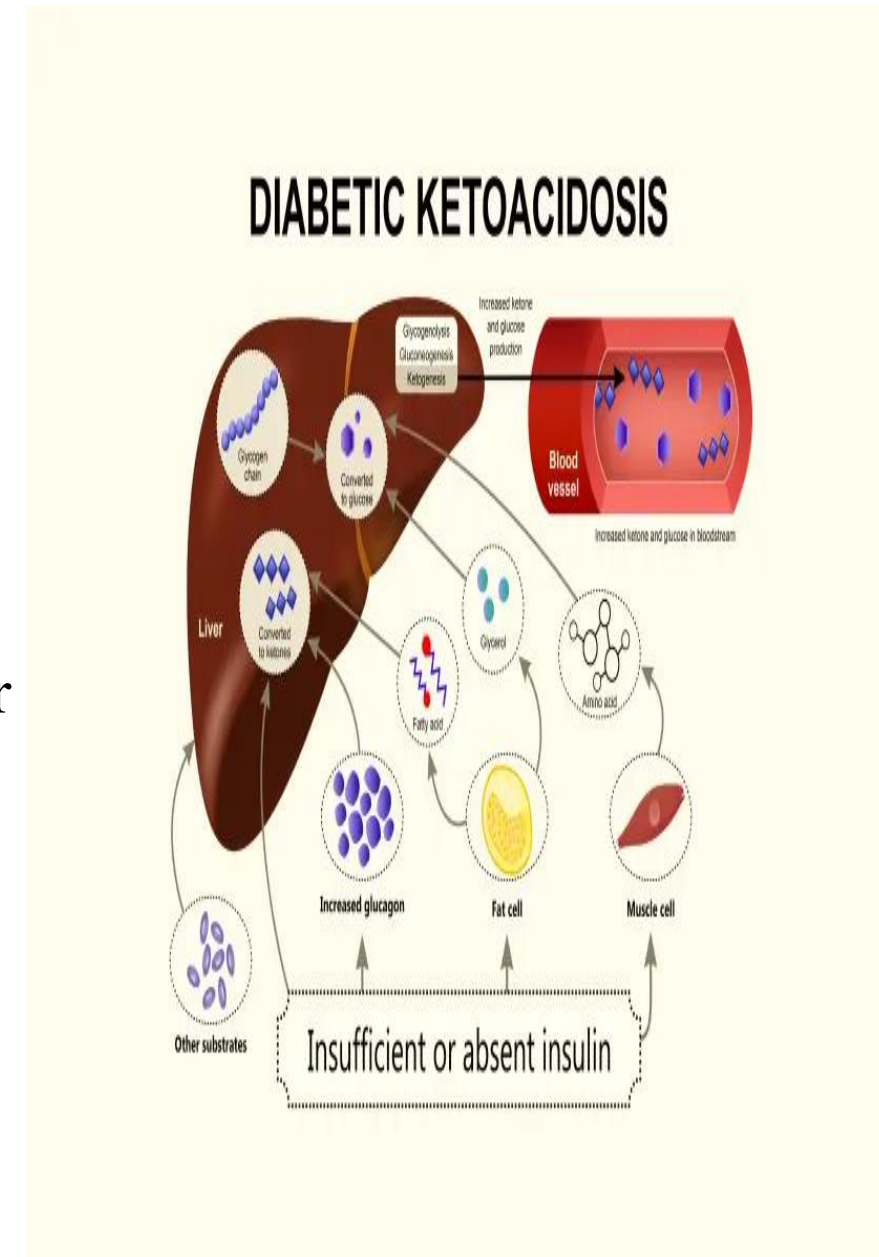
Etiology of DKA

- * **Infection** is a major reason that diabetic patients progress to DKA.
- * Changes in the type of insulin, dosage, or increased metabolic demand can precipitate DKA in **individuals with type I Diabetes Mellitus**.
- * Surgery, infection, and trauma because require an increase in insulin intake.
- * In young persons with DM, psychological problems combined with eating disorders
- * DKA also occurs with acute pancreatitis.
- * Non diabetes causes of DKA are **starvation and alcoholism**.

Pathophysiology of DKA

Insulin Deficiency

- Insulin is the metabolic key to the transfer of glucose from the blood stream into the cell, where it can be used for energy.
- Without insulin, glucose remains in the blood stream, and cells are deprived of their energy source.
- When the body lacks insulin and cannot use carbohydrate for energy, the liver convert its glycogen stores back to glucose or breaking down stored glucose (Glycogenolysis).
- These responses worsen the situation by raising the blood glucose level even higher.



Hyperglycemia

Hyperglycemia causes an osmotic diuresis. High extracellular glucose levels produce an osmotic gradient between the interacellular and extracellular spaces, causing fluid to move out of the cells.



Fluid Volume Deficit

When the serum glucose exceeds, the renal threshold, it cannot be reabsorbed at the renal tubule then glucose is lost through the kidneys and spills into the urine (Glycosuria).

Polyuria and glycosuria

occur as a result of the osmotic diuresis. As a result, large volumes of water, along with sodium, potassium, and phosphorus, are excreted in the urine, causing **a fluid volume deficit**. Osmotic diuresis increases serum osmolality and the blood becomes hyperosmolar.

Hyperosmolar cause further fluid shift from the intracellular to the extracellular space results in cellular dehydration.

Ketoacidosis

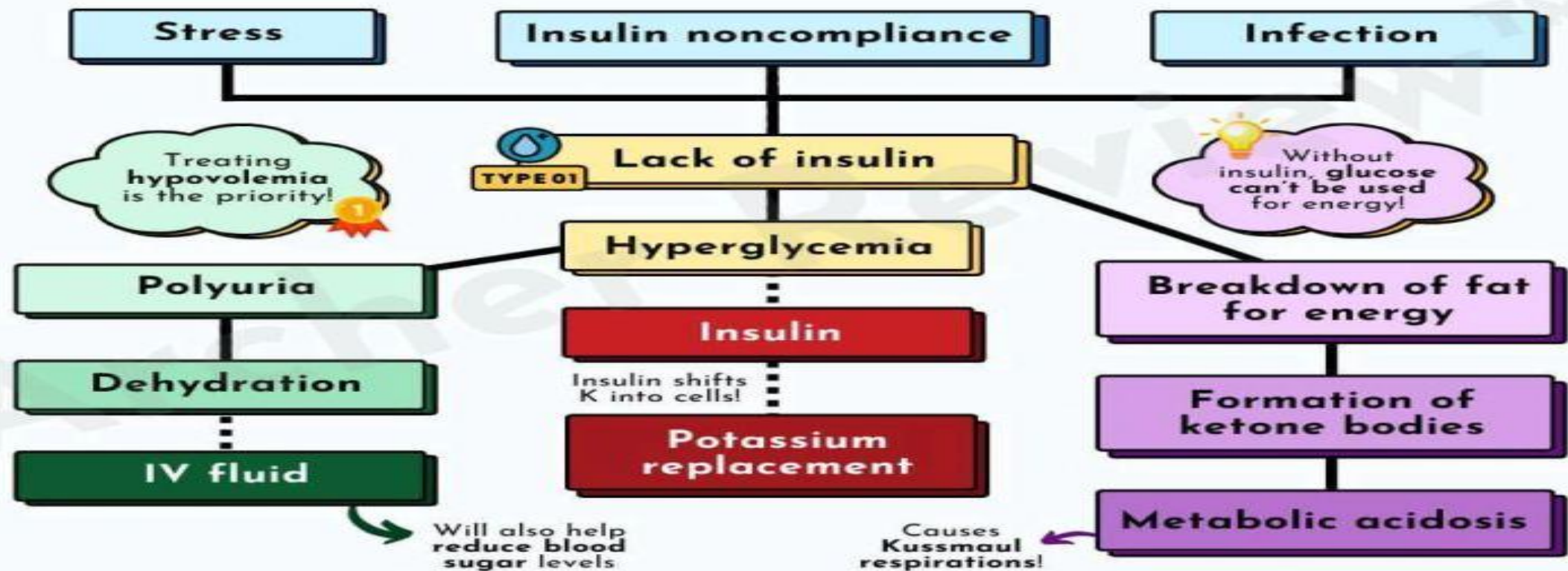
- ❖ The body begins to use fat and protein stores for energy.
- ❖ Excessive amount of fatty acid are mobilized from adipose tissue cells and transported to the liver.
- ❖ The liver in turn, increases the production of ketone bodies for catabolism by the muscle.
- ❖ As fat metabolism increases, the liver produces too many ketone bodies.
- ❖ Ketone bodies accumulate in the blood (ketosis) and excreted in the urine (ketouria).
- ❖ Accumulation of ketone bodies causes metabolic acidosis. Sever acidosis may cause the diabetic patient to lose consciousness.

Acid-Base disturbance

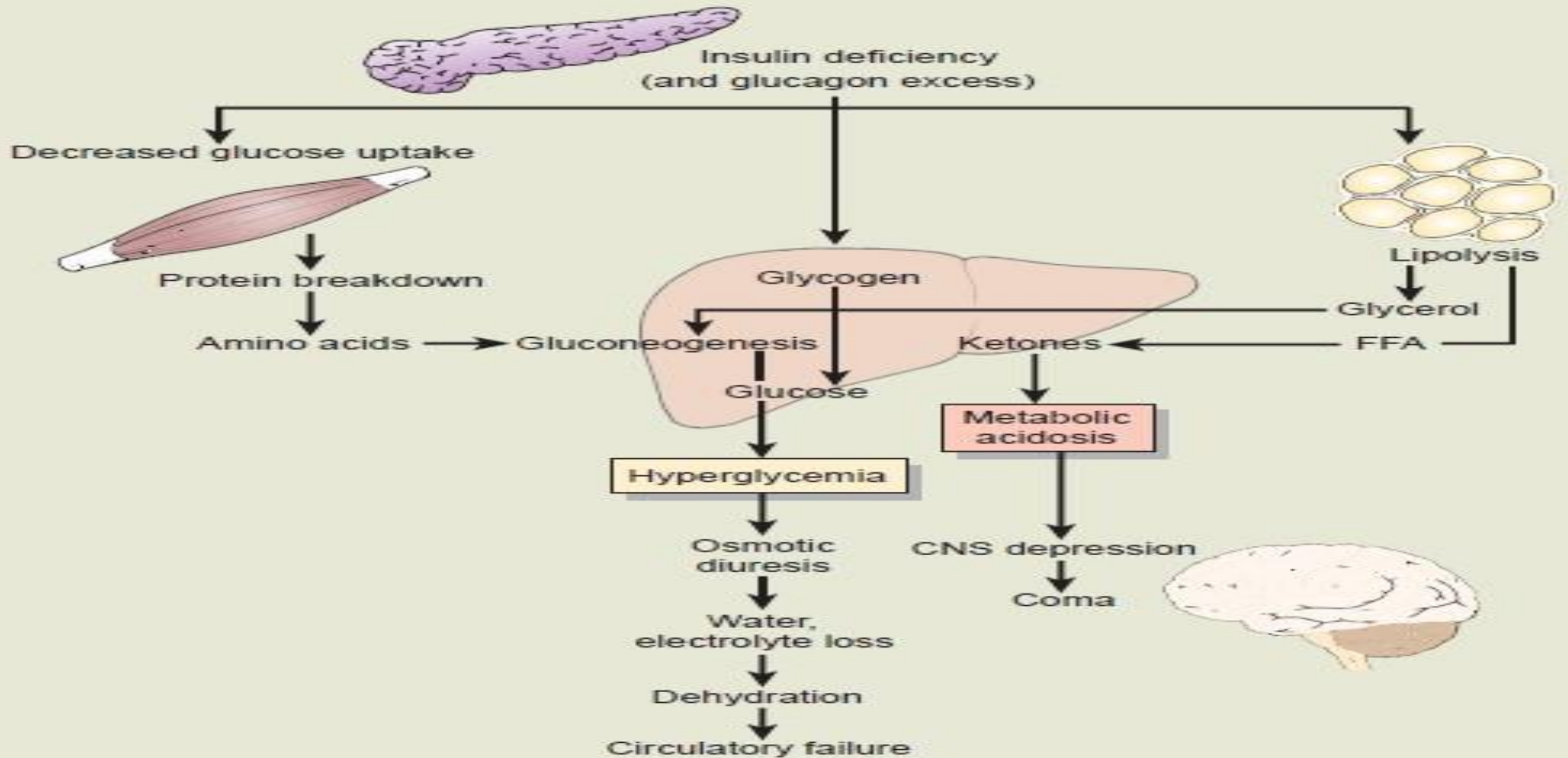
Increase ketones , decrease pH and decrease bicarbonate cause **metabolic acidosis**.

- ✓ Breath becomes deep and rapid (Kussmaul Respiration) to release carbonic acid in the form of Carbon Dioxide (CO₂).
- ✓ Acetone is exhaled, giving the breath its characteristic of fruity odor.

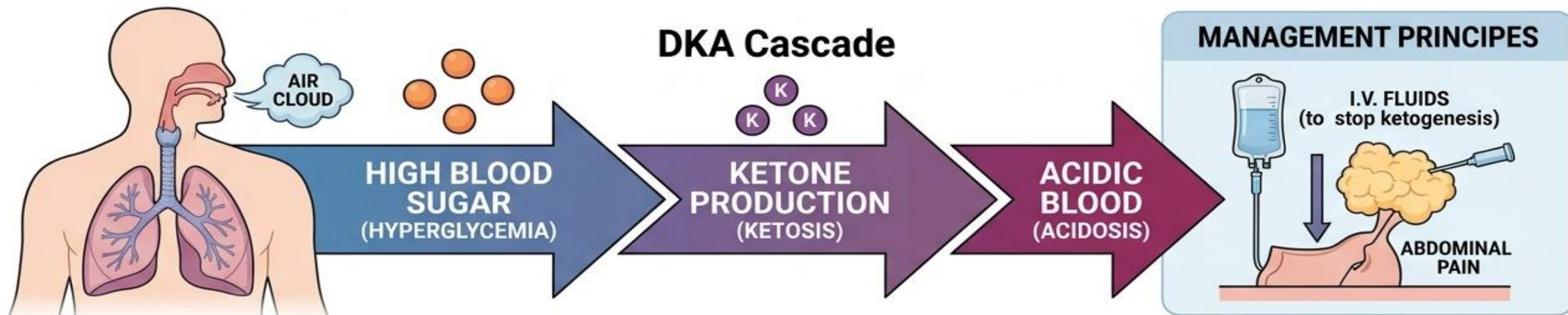
Diabetic Ketoacidosis (DKA) Pathophysiology and Treatment



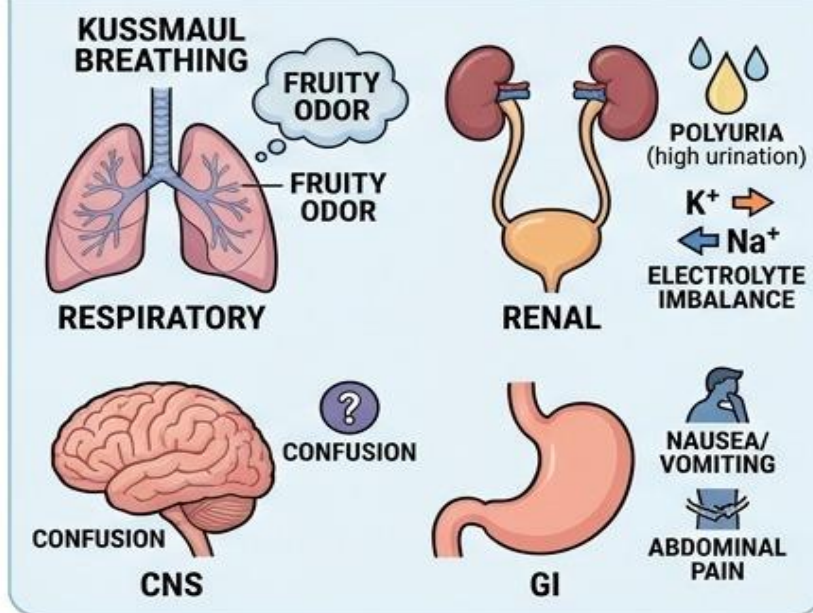
Pathophysiology DKA



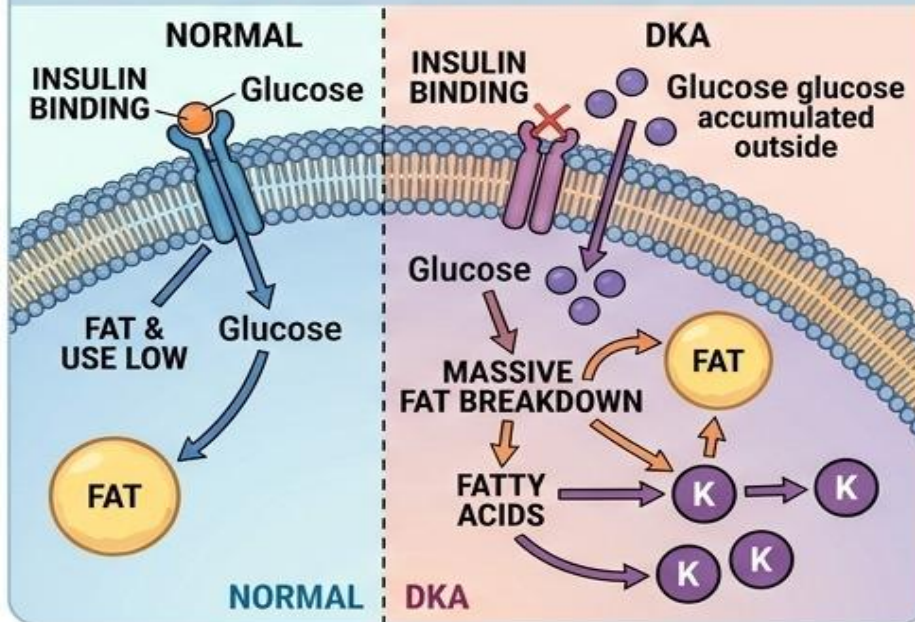
UNDERSTANDING DIABETIC KETOACIDOSIS (DKA)



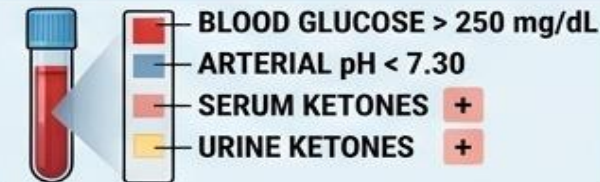
SYSTEMIC EFFECTS



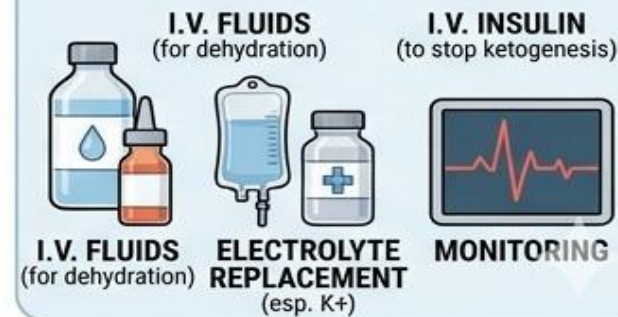
CELLULAR MECHANISM



LABORATORY DIAGNOSIS



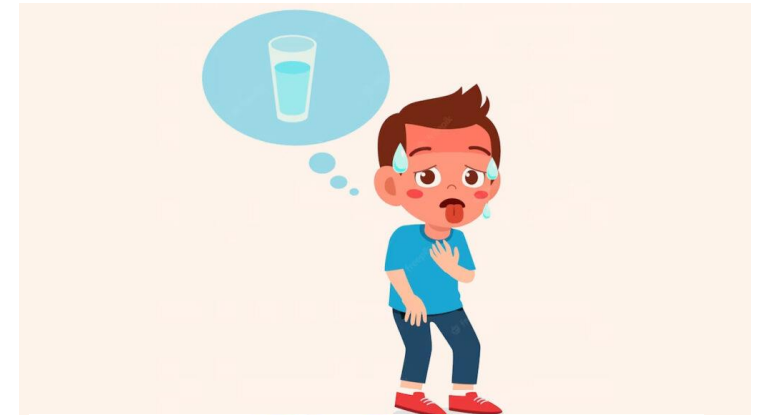
MANAGEMENT PRINCIPLES



Clinical Manifestation

Patient complaints of

- Dry mouth
- Malaise
- Headache
- Polyuria (excessive urination)
- Polydipsia (excessive thirst)
- Polyphagia (excessive hunger)
- Dehydration (dry skin, hypotension, weakness)



Polyphagia



Clinical Manifestation

*** Ketoacidosis**

- * Air hunger
 - Acetone breathe odor
- * Deep and rapid respirations
 - Nausea, vomiting
- * Extreme fatigue, dehydration, and weight loss follow.

*** Central nervous system**

- * Depression
- * Changes in the level of consciousness, can lead quickly to coma.

How to diagnosis DKA



The physical examination

- * Reveals evidence of **dehydration**, including: flushed dry skin and dry buccal membranes
 - ***Skin turgor** : that takes longer than 3 seconds to return to its original position after the skin has been lifted.
 - ***Sunken eyeballs**, resulting from lack of fluid in the interstitial of the eyeball, are observed
 - ***Tachycardia and hypotension** may signal profound fluid losses.
 - ***Kussmaul respirations** are present, and the fruity odor of acetone may be detected.

Laboratory Studies

- * Hyperglycemia
- * Urine ketones
- * If **a blood gas sample** is obtained, it can confirm the acid-base imbalance.

The venous blood chemistry panel include: low CO₂ in the presence of uncompensated metabolic acidosis.

- * **Serum sodium** may be low because water moves from the intracellular space into the extracellular (vascular) space.
- * **The serum potassium** level may be elevated, as potassium moves from inside the cell to the extracellular space

Management DKA

Treatment goals for DKA include:

- Reverse Dehydration
- Replace Insulin
- Reverse ketoacidosis
- Replenish electrolyte

Reversing Dehydration

- ❖ Patients with DKA are dehydrated and may have lost 5% to 10% of body weight of fluid.
- ❖ Aggressive IV fluid replacement is provided to rehydrate intracellular and extracellular compartment and prevent circulatory collapse.
- ❖ IV isotonic normal saline (0.9% Sodium Chloride (Na Cl) is infused to replenish the vascular deficit and to reverse hypotension.

Reversing Dehydration

- If the serum osmolality is elevated and serum sodium is high (Hypernatremia), infusion of hypotonic solution as Na Cl (0.45%) follow the initial saline replacement.
- The replacement infusion includes 20 to 30 mEq of potassium per liter to restore the intracellular potassium.
- Careful attention must be paid to patients with kidney failure or cardiopulmonary disease during fluids replacement to avoid fluids overload.
- After blood glucose level decrease to 200 mg/dl, the infusion solution is changed to a 50/50 mix of hypotonic saline and 5% dextrose. Dextrose prevents unexpected hypoglycemia.

Insulin Administration

- ❖ Administer an initial IV bolus of regular insulin at 0.1 unit for each kilogram of body weight. A continuous infusion of regular insulin at 0.1 unit /kg/hour with IV infusion (if patient weight 70 kg , infusion of insulin 7 unit per hour).
- ❖ Blood glucose level should be checked every hour to assess plasma glucose level and its decrease of between 50 to 70 mg/dl per hour.
- ❖ The insulin rate may be decreased to 0.05 to 0.1 unit /kg/hour (usually 3 to 6 units for adult who are receiving continuous IV insulin), after the glucose level is decreased to 200 mg/dl, the acidosis has been corrected and rehydration has been achieved.

Reversing Ketoacidosis

- Adequate hydration and insulin replacement usually correct acidosis but many patients may require bicarbonate replacement if pH value is less than 7.10.
- The bicarbonate deficit can be calculated and replaced IV over several hours to raise the level at least to the 10 to 12 mEq/L range.
- Sodium bicarbonate should be administered by slow IV infusion over several hours. It is administered as a bolus injection only in the case of cardiac arrest.
- Sodium bicarbonate administration can cause a rapid reduction in plasma potassium concentration and sodium overload.

Replenishing Electrolytes

- ❑ Low serum potassium (**Hypokalemia**) occurs as insulin promotes the return of potassium into the cell and metabolic acidosis is reversed.
- ❑ The potassium level must be 3.3 mEq or greater before administering insulin. Potassium chloride is administered as soon as the potassium level falls below the normal. Potassium is usually replaced at concentrations of 20 to 40 mEq/L of IV fluid, depending on the serum potassium level.
- ❑ Phosphate replacement is recommended if phosphate level is less than 1mg/dl.
- ❑ Patients who are receiving IV phosphate therapy should be watched carefully for signs of tetany (tingling around the mouth or in the hands, neuromuscular irritability, carpopedal spasm, or even seizures). Tetany can occur because phosphate lowers the level of circulating calcium.

Reestablishing Metabolic Function

- Gastric motility is greatly impaired in DKA ,Gastric distention with dark, hemo-positive fluid and vomiting is common. Abdominal pain, tenderness, and a paralytic ileus may also occur due to DKA.
 - The patient may need a nasogastric tube to decompress the stomach. This increases comfort and decreases the risk for aspiration.
 - Patients should not eat or drink in this phase of illness. Ice chips may decrease thirst.
- Later, when distention lessens and motility returns, oral intake begins in order to provide the complex nutritional requirements for recovery

Nursing Management Goals

- ✓ Administering fluids, insulin, and electrolytes
- ✓ Monitoring response to therapy
- ✓ Surveillance for complications
- ✓ Providing patient education

Administering Fluids, Insulin, and Electrolytes

- ❖ Rapid intravenous fluid replacement requires the use of a volumetric pump.
- ❖ Insulin is administered intravenously to patients who are severely dehydrated or have poor peripheral perfusion, to ensure effective absorption.
- ❖ Patients with DKA are kept on NPO status (nothing by mouth) until the hyperglycemia is under control.
- ❖ Assess the patient's response and the laboratory data for changes relating to blood glucose levels.
- ❖ The critical care nurse is responsible for monitoring the rate of plasma glucose decline in response to insulin.

Administering Fluids, Insulin, and Electrolytes

- ❖ The goal is to achieve a fall in glucose level of approximately 50 to 75 mg/dL hourly.
- ❖ Monitoring blood glucose, potassium, and often blood gases on an hourly basis is considerable.
- ❖ When the blood glucose level falls to 200 mg/dL, a 5% dextrose solution (D5W) with 0.45% Na Cl solution is infused to prevent hypoglycemia.

❖ The regular insulin drip is not discontinued until the ketoacidosis subsides, as identified by **absence of ketones and a normal pH by arterial or venous blood gas** analysis.

❖ **Insulin** is given subcutaneously after **glucose levels, dehydration, hypotension, and acid-base balance are normalized** and the patient is in stable condition and taking an oral diet.

Monitoring Response to Therapy

- ❖ Assess occurrence of hypoglycemia during the transition period.
- ❖ Discontinuing the insulin and intravenous dextrose infusion, a patient receives a subcutaneous dose of insulin and is expected to eat a meal.
- ❖ If the patient is unable to eat an adequate amount, hypoglycemia results from the administration of subcutaneous insulin without adequate glucose.

Surveillance for Complications

- * Fluid volume overload
- * Hypoglycemia
- * Hyperkalemia
- * Hyponatremia
- * Cerebral edema
- * Infection

Nursing Diagnosis *Diabetic Ketoacidosis*

- Decreased Cardiac Output related to alterations in preload,
- Deficient Fluid Volume related to absolute loss,
- Anxiety related to threat to biologic, psychologic, and social integrity
- Disturbed Body Image related to functional dependence on life- sustaining technology,
- Ineffective Coping related to situational crisis and personal vulnerability
- Powerlessness related to lack of control progression,
- Deficient over current situation or disease progression,
- Deficient Knowledge related to lack of previous exposure to information

Comparison between Diabetic and Hyperglycemic Hyperosmolar Syndrome

Characteristics and Laboratory Tests	DKA	HHS
Cause	Insufficient exogenous glucose for glucose needs	Insufficientexogenous/ endogenous insulin for glucose needs
	Sudden (hours)	Slow, insidious (days, weeks)
Onset Precipitating factors	Noncompliance with type 1 diabetes therapy, illness, surgery, decreased activity	Recent acute illness in older patient; therapeutic procedures

Comparison between Diabetic and Hyperglycemic Hyperosmolar Syndrome

Characteristics and Laboratory Tests	DKA	HHS
Mortality (%) Patients affected	9-14	10-50
Clinical manifestations	Patients with type I diabetes Dry mouth, polydipsia, polyuria, polyphagia, dehydration, dry skin, hypotension, weakness Ketoacidosis, air hunger, acetone breath odor, respirations deep and rapid, nausea, vomiting	Patients with type II diabetes Mental confusion, tachycardia, changes in level of consciousness No ketosis, no breath odor, respirations rapid and shallow, usually mild nausea/vomiting

Comparison between Diabetic and Hyperglycemic Hyperosmolar Syndrome

Laboratory tests	DKA	HHS
Glucose (mg/dL)	300-800	600-2000
Ketones pH Osmolality (mOsm/L)	Strongly positive	Normal or mildly elevated Normal
Sodium	<7.3	>350
Potassium (K ⁺) Bicarbonate	<350	Normal or elevated
Phosphorus	Normal or low	Low, normal, or elevated Normal
	Normal, low, or elevated (total body K ⁺ depleted)	Low, normal, or elevated (may decrease after insulin therapy)
	<15 mEq/L	
	Low, normal, or elevated (may decrease after insulin therapy)	
Urine acetone	Strong	Absent or mild

MCQ about DKA

Which condition is identified as a major precipitating factor that causes diabetic patients to progress into DKA?

- A. Excessive exercise without dietary adjustment.
- B. Acute infection.
- C. High protein intake.
- D. Over-hydration with hypotonic fluids.

When managing DKA, at what blood glucose level should the IV solution be changed to include 5% Dextrose?

- A. 300 *mg/dL*
- B. 100 *mg/dL*
- C. 500 *mg/dL*
- D. 200 *mg/dL*

Thank
you

